LOPA Study Manager – Quick Guide

Layer of protection analysis linked to the HAZOP · part of the Process Safety Analysis Toolkit

1 · What it does

A LOPA checks whether the independent protection layers (IPLs) on one scenario bring its frequency below the target. One scenario is **one initiating event leading to one consequence**. The tool takes the scenarios from the HAZOP, calculates each one for five receptors (safety, environment, asset, reputation, community), records the adjudication of every IPL credit and flags common crediting errors.

HAZOP: IPL ticked LOPA → LOPA scenario → Initiating event × enabling conditions × conditional modifiers × IPL PFDs → Compare with the target → Gap, SIL, ALARP → Results back to the HAZOP

2 · Workflow

1. **In the HAZOP Study Manager**, set each protection that should be credited to **IPL** and tick **LOPA**. Save the HAZOP study file.
2. **Risk criteria** – choose *explicit* (frequencies against the target maximum event likelihood, TMEL) or *implicit* (whole credits against the credits required). Enter the client's targets for each receptor and severity. The defaults are typical examples only.
3. **File ▸ Import HAZOP study** – every initiating cause of a flagged deviation becomes a scenario (L-01, L-02 …) with the severity by receptor, the IPLs ticked LOPA and the other safeguards (listed, not credited). Import again after HAZOP changes: the HAZOP text updates and your LOPA entries stay.
4. **Scenario worksheet** – work through the layers (section 3), then adjudicate each IPL (section 4).
5. **Rules check** – clear every finding: fix the scenario or record why the credit stands.
6. **Reports & export** – report, Excel, bow tie. In the HAZOP, **File ▸ Import LOPA results** puts the status and result on the LOPA study list.

3 · The scenario worksheet

Layer	What to enter
Initiating event	Frequency per year and its basis. Pick from the data library (CCPS LOPA 2001 Table 5.1, CCPS 2015 data tables) or use site data.
Enabling conditions	A condition that must be present, e.g. the fraction of the year in a campaign. Tick **Time at risk** and answer whether a failure outside that period would be found before it starts – if not, it is not an enabling condition.
Conditional modifiers	Probability of ignition, of a person being present, of injury. Tick the receptors each one applies to (presence: safety only). Library: CCPS 2014 guidelines.

IPLs	PFD, type and basis for each IPL. Tick **Proposed** for a new IPL recommended to close a gap – it shows in a separate “with proposed IPLs” result.
Result	For each receptor: the mitigated frequency (or credits), the target, the risk reduction still needed (RRF) and the result: *Meets target*, *Gap ≤ 10× – one more IPL or SIL 1*, *SIL 1 / 2 / 3 needed*, or *Above SIL 3 – change the design*.
ALARP	A result that meets the target but is within the ALARP band (two decades by default) asks for the further measures considered, the decision and its justification.

> **Implicit LOPA** rounds every factor down to whole credits (0.05 → 1 credit, 6 /yr → −1). It is more conservative than the explicit calculation – in the example, L-01 needs SIL 1 explicitly and SIL 3 implicitly.

4 · Adjudicating an IPL credit

Open the panel under each IPL. Answer the seven tests – **specific, independent, dependable, auditable, access & change control, in time, cyber-secure** (cannot be defeated by a cyber attack). Record the evidence (proof-test records, SRS, inspection reports, procedure), the reviewer and the status:

Claimed → **Under review** → **Accepted / Accepted with conditions / Rejected**. An accepted PFD can be set higher than the PFD claimed (less credit). A rejected IPL is never counted. Every change goes in the **Adjudication log**.

On *Risk criteria*, choose which IPLs count: **all claimed** (except rejected) while the study runs, or **accepted only** for the final result.

5 · Rules check

The check flags – the reviewer judges. Each finding names its source. Mark a finding **Justified** with the reason, or fix the scenario.

Finding	Source
A failed IPL test; the same IPL credited twice	IPL criteria, CCPS LOPA (2001)
BPCS credit below PFD 0.1; a BPCS loop failure as the cause with a BPCS IPL	IEC 61511; CCPS IPS (2007) §3.5
Operator response below 0.1, or not shown to fit the time available	CCPS IPS (2007) Table B.2; Marszal & Scharpf Table 9.1
Relief device not confirmed as sized for the scenario	CCPS IPS Table B.4; IChemE LPB 265
Administrative IPL; fire & gas system above RRF 10	CCPS IPS Tables B.1, B.5
Time at risk on an unrevealed failure; operating hours on a human-error initiating event	CCPS ECCM (2014) §2.3; IChemE LPB 265
Presence below 0.1; probability of explosion; ignition credit for a classified area only	CCPS ECCM Table 3.4, §3.3; Marszal & Scharpf §9.4

ALARP decision not recorded	UK HSE Tolerability of Risk

6 · Registers and outputs

- **IPLs credited** – every IPL with the scenarios that rely on it; a PFD that differs between scenarios is flagged.
- **SCAI register** – SIFs, BPCS functions and alarms credited, with the SIL needed (from the PFD credited), set point, proof-test interval, response time and owner – the input to SIL verification in the Fault Tree & PFD Calculator.
- **Export bow tie** – scenarios that share a top event become threats with their IPLs as barriers; open the file in Bow Tie Analysis.
- **Report and Excel** – study, criteria, scenarios, results by receptor, IPL adjudication, rules check, SCAI register, ALARP and the log.

7 · Example and saving

File ▸ Load example study opens the hot DRI conveyor LOPA (L-01 stoppage under load, L-02 over-temperature DRI, L-03 N2 asphyxiation) – the same scenario as the HAZOP, fault tree, event tree and bow tie examples. Values are illustrative, not design data. Save with **File ▸ Save LOPA as…** to a `.lopa.json` file (Chrome / Edge write every change to it); use **Download backup** as a second copy.

> The data library values are screening values. “Verify” items are not checked against a book. Use site, vendor or verified data where you have it, and the client's own criteria and credit tables.